

eyetools: an R package for open-source analysis of eye data

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Abstract

Eye data analysis has become an integral part of research in both academic and commercial settings. In the behavioural sciences there is an ever increasing desire and need to shift analysis from proprietary software to open source alternatives. Here we introduce eyetools, an R package that provides an open source means of conducting eye data analysis. eyetools is aimed at researchers who may not have experience programming their own eye data analysis routines. The package provides a number of functions that facilitate key steps in the pipeline of eye data analysis, from basic data manipulation, extraction of fixations and saccades, to trial level summaries and visualisations. In this article we introduce these core features of the package and provide a step-by-step guide which will enable researchers to find open source solutions to their eye data analysis. We discuss the ways in which we believe eyetools might be expanded in the future and how it may provide a platform for collaborative work on eye data analysis.

Keywords: software; R; eye-tracking; fixations; saccades; areas-of-interest

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Eye-tracking is now an established and widely used tool that provides powerful measurements of human behaviour. Its application across academic research is far-reaching, with major importance to the fields of computer science and AI, economics, psychology, and even having influence on art and design. Beyond purely academic application, eye-tracking is widely used in commercial fields where it can provide insights into consumer behaviour, helping to shape product design and marketing.

By recording the movement of an individual's gaze during research studies, users can quantify where and how long individual's look at particular regions of space (usually with a focus on stimuli presented on a 2D screen, but also within 3D space). Eye-tracking provides a rich stream of continuous data which can offer powerful insights into real-time cognitive processing. Undoubtedly, Psychology (and related broad areas of research such as vision science, linguistics, etc) is perhaps the scientific discipline which has seen the most substantial adoption of eye-data research, where eye-tracking systems are now commonplace in centres of academic research. Such data allow researchers to inspect the interplay of cognitive processes such as attention, memory, decision making, and reading, with high temporal precision ([Beesley et al., 2019](#)).

While there are abundant uses and benefits of collecting eye-movement data, the collection of such data has many implications for the eventual analysis work that is undertaken by the researcher. The continual stream of recording can lead to an overwhelming amount of raw data: modern eye-trackers can record data at 1000 Hz and above, which results in 3.6 million samples taken per hour. The continual nature of the data also leads to a wealth of choice in the manner in which it is analysed. As such, the provision of suitable computational software for data reduction and processing is an important part of eye-tracking research. The companies behind eye-tracking devices offer licensed software that will perform many of the common steps for eye-data analysis. However, there are several disadvantages to using such proprietary software in a research context. Firstly, the software will typically have an ongoing license cost for continual use. Secondly, the algorithms driving the operations within such software are not readily available for inspection or

adaptation for new purposes. These significant constraints mean that the use of proprietary analysis software will lead to a failure to meet the basic open-science principle of analysis reproduction, for example as set out by the UK Reproducibility Network: “We expect researchers to... make their research methods, software, outputs and data open, and available at the earliest possible point...The reproducibility of both research methods and research results ...is critical to research in certain contexts, particularly in the experimental sciences with a quantitative focus...”.

In the current article we introduce a new toolkit for eye-data processing and analysis called “*eyetools*”, which takes the form of an R package. R packages (like R itself) are free to use and are therefore available for users across the world. The package provides a (growing) number of functions that provide an efficient and effective means to conduct eye-data analysis on data from typical laboratory 2D eye-tracking systems (i.e., Tobii Spectrum and similar; EyeLink 1000 Plus and similar), for tasks that employ static 2D displays of stimuli (there are no functions for smooth pursuit analysis). *eyetools* is built with academic researchers in the psychological sciences in mind, though there is no reason why the package would not be effective more generally. The functions within the package reflect steps in a comprehensive analysis pipeline, taking the user from initial handling of raw eye data, to summarising data for each period of a procedure, to the visualisation of the data in plots. Since the pipeline is contained within the one package, it is not reliant on external software, which enables easy reproducibility of any analyses. Importantly, the functions are simple to use, ensuring that the package will be beneficial for researchers who are unfamiliar with eye data analysis. It should also appeal to researchers accustomed to working with eye data in other environments who wish to transfer to working in R.

eyetools is, of course, not the only package in R that allows users to work with eye data. We conducted a survey of the available packages on CRAN (The Comprehensive R Archive Network), by searching the list of available packages (https://cran.r-project.org/web/packages/available_packages_by_name.html) for the terms “eye” (which covers terms “eye-tracking” and “eyetracking”), “gaze”, “saccades” and “fixation”. We

considered packages that were designed for work within the psychological sciences¹, which identified 12 packages that offer relevant functions for the analysis of eye data, shown in Table 1. *eyeTracker*, *eyelinker*, and *eyelinkReader*, all offer functionality for data collected specifically from ‘EyeLink’ trackers (SR Research). The *emov* package offers a limited set of functions and is primarily designed for fixation detection, using the same dispersion method employed in *eyetools*. *eyetrackingR* and *kollaR* are perhaps the most comprehensive alternative packages and having some similarity to *eyetools* in their broad use-case. *eyetrackingR* offers a large suite of functionality and, like *eyetools*, can be applied across the entire pipeline. It has functions for assessing missing data, analysing data over time and visualising data with various plotting functions. It does not feature algorithms for the detection of events such as saccades or fixations. *kollaR* is hardware-agnostic and offers functions for data processing, fixation detection and some plotting features. Several packages offer specific use-cases: *saccadr* is a package that can be used to extract saccades from raw data, and has functionality to convert binocular data into monocular data; *fpa* offers a particular spatio-temporal analysis and associated plots; *gazePath* offers detection of fixations and saccades, in addition to an optional GUI. The packages *eyeris* and *Pupillometry* are focussed on the analysis of pupillometry data, which is not a focus of the *eyetools* package (at present).

In this tutorial we demonstrate the pipeline of analysis functions within *eyetools*. The package has been designed to be simple to use by someone with basic knowledge of data handling and analysis in R. We envisage that this package can be used by (for example) first-year PhD students or early career researchers with minimal experience working in R. Users will need to understand basic R functionality like how to perform simple operations on data objects. Our hope is that the package will enable researchers who haven’t previously engaged with eye-tracking to do so in an open-source and reproducible manner. We first describe the basic installation process and

¹ We did not present or summarise here packages that were designed for work in ophthalmology research. There are also several packages that are not listed on CRAN but can be installed using alternative methods. Of note is the package *gazeR* which, like *eyetools* and *kollaR*, features a comprehensive array of functions for work with eye data.

Table 1*Properties of CRAN packages designed to work with eye-data for the psychological sciences*

Package	Hardware-agnostic	Data Import	Data processing	Fixation detection	Saccade detection	Pupillometry	Plotting
emov	✓		✓	✓	✓		
eyelinker		Eyelink					
eyelinkReader		EyeLink		✓	✓		✓
eyeris	✓	EyeLink	✓			✓	✓
eyetools	✓	Tobii	✓	✓	✓		✓
eyetrackingR	✓		✓				✓
eyeTrackR		EyeLink	✓				
fpa	✓		✓				✓
gazePath	✓			✓			
kollaR	✓		✓	✓			✓
PupillometryR			✓			✓	✓
saccadr	✓			✓			

Note. Hardware-agnostic: whether designed for use only with a proprietary data format; Data Import: has functions for direct import of data from eye-tracking hardware; Data processing: basic processing functions, e.g. interpolation, smoothing. Fixation/Saccade detection: functions for identifying events from raw data; Pupillometry: functions for pupillometry data; Plotting: functions to plot raw or processed data

then the preparation of data into an *eyetools*-friendly format. We then describe the algorithms that can be used to extract the key characteristics of fixations and saccades. We then show the algorithms that can extract critical trial level patterns in behaviour, such as time on areas of interest and sequences of eye movements, as well as methods to visualise the data.

Installing *eyetools*

eyetools is available on CRAN and can be installed with the command `install.packages("eyetools")`. Instructions for installing development versions can be found at the package repository: <https://github.com/tombeesley/eyetools/>. Once installed, the package can be loaded into R with the command `library(eyetools)`.

Preparing data for *eyetools*

Since there is a wide range of eye tracking hardware available for researchers to use, *eyetools* currently offers limited functionality for converting raw data from specific hardware. The `hdf5_to_dataframe()` function is designed to work with output from PsychoPy experiments connected to modern Tobii hardware. This function takes the default raw data format and converts it into a simplified raw data format suitable for *eyetools*.

The *eyetools* package has been developed primarily with the analysis of experimental psychology data in mind. It is designed for, and tested with, data that has been collected from standard static 2D displays.

In the prototypical experimental psychology task, static displays of stimuli will be presented for a period of time before the presentation terminates and an inter-trial-interval occurs. As such, many of the functions expect a “trial” variable in the data, such that the algorithms will operate over multiple trials and produce output that retains this trial information. Similarly, data in psychology experiments tends to come from multiple participants, and to facilitate analysis, a “pID” column is required (even if data from only a single participant is used). This means that the user can avoid having to generate additional programming steps to analyse and combine the data from multiple participants. It is quite typical in psychology experiments for there to be multiple defined periods within a trial, such as fixation, different periods of stimulus presentation, feedback presentation, inter-trial-interval, etc. Since each experimental paradigm will have a particular arrangement of such periods, *eyetools* does not interpret these changes automatically, and so it is necessary to first select the data for the period or periods that are of interest for analysis. Analysis on each period would be conducted separately using the functions in *eyetools*.

The starting point for the analysis pipeline is the preparation of the raw eye data, which will consist of recorded samples from the eye-tracker, with each row in the data reflecting a single time-stamped recording. If the eye-tracker is set at 1000Hz, then consecutive recordings will be 1 millisecond of time apart; at 300Hz, the recordings are $3.\bar{3}$ milliseconds apart. The only requirement for the time column is that the values reflect a consistent and increasing set of values. There is no need to specify the sampling rate, since *eyetools* functions will calculate this automatically. For the majority of functions that process raw data, *eyetools* expects raw data to have the following columns:

- x = horizontal spatial coordinate of the estimated eye position
- y = vertical spatial coordinate of the estimated eye position
- time = timestamp of the recording
- trial = the index of the current trial in the data
- pID = the unique identifier for the data from each participant

The first four columns should be set as type numeric, while “pID” can be numeric, character, or factor. The order of the columns is not important. Missing values in the x and y columns of the raw data must be expressed as “NA”.

While *eyetools* works on monocular data (data from a single eye, i.e., a series of X and Y coordinates) in the main, many eyetrackers will output binocular data (data from two eyes, i.e., an X and Y coordinate for each eye). In such cases, since the primary aim of our analyses is the estimation of the spatial coordinates of gaze, the function `combine_eyes()` should be used to combine the data to form a set of monocular data. This function takes raw data with coordinates for each eye (i.e., `left_x`, `right_x`, `left_y`, `right_y`), and converts the data into single x and y coordinates. By default, the function does this by taking an average of the coordinates from the two eyes of each timestamp, but it is also possible to select data from the eye with the lowest

proportion of missing samples. This function returns data that has x and y variables in place of the left_* and right_* variables.

```
head(HCL,4) # first 4 rows of the built-in data
```

```
# A tibble: 4 x 7
```

```
  pID    time left_x left_y right_x right_y trial
  <chr> <dbl>  <dbl>  <dbl>   <dbl>   <dbl> <dbl>
1 118      0   909.   826.   1003.    808.     1
2 118      3   912.   829.   1001.    812.     1
3 118      7   912.   826.   1010.    813.     1
4 118     10   908.   824.   1006.    807.     1
```

```
data <- combine_eyes(HCL) # create monocular data
```

```
head(data, 4) # first 4 rows of the monocular data
```

```
  pID time trial      x      y
1 118   0     1 955.8583 816.5646
2 118   3     1 956.5178 820.6221
3 118   7     1 960.7383 819.7616
4 118  10     1 956.9727 815.3331
```

Repairing missing data and smoothing data

Despite the best efforts of the researcher, there are occasional failures in the accurate recording of the eye position during data collection (e.g., blinks). This results in missing data within the stream of samples, which must be represented in eyetools as NA values for the x and y coordinates. To mitigate the impact of missing data on further analysis, the `interpolate()` function can estimate the missing gaze data, based upon the eye coordinates before and after the

missing data, and perform a repair. The default method of linear interpolation (“approx”) replaces missing values with a line of constant slope and evenly spaced coordinates that bridge between the existing data (alternatively a cubic “spline” method can be used to apply a curved path between the existing datapoints).

```
data <- interpolate(data,
                    method = "approx")
```

When using `interpolate()`, a report can be requested on the proportion of missing data that has been replaced. This parameter changes the output format of the function, and returns a list of both the data and the report. The report alone can be accessed in the following way:

```
interpolate(data,
            method = "approx",
            report = TRUE)[[2]]
```

	pID	missing_perc_before	missing_perc_after
1	118	0.02314313	0.001157156
2	119	0.01214128	0.000000000

As shown, not all missing data has been replaced, since there are certain periods in which the missing data span a period longer than the default setting of the “maxgap” parameter, which is 150 ms.

Once interpolation has been performed, a common step is to smooth the eye data to minimise the effect of measurement error on the data. The function `smoother()` reduces the noise in the data by applying a moving average function. The degree of smoothing can be specified, and a plot can be generated (using data from a randomly selected trial) to observe how well the smoothed data fits the raw data.

```
data <- smoother(data,  
                  span = .02,  
                  plot = TRUE)
```

Working with *eyetools*

Having explained these rudimentary steps of getting the data ready for the main analysis, we will now describe the core functions available in the latest version of *eyetools*. For illustration, *eyetools* has a built in data set that meets the required format. The data set consists of data from two participants from a few trials of a human causal learning study (Beesley et al., 2015). The nature of this experiment is largely unimportant for the current purposes, but for clarity, the data were collected from the decision period of the procedure, where two rectangular cue stimuli were presented in the top half of the screen, one on the left side and one on the right side. Two smaller response options were presented centrally in the lower half of the screen, one above the other. Participants simply had to look at the cues and choose a response. The raw eye data can be accessed by calling HCL, the “behavioural data” (trial events, reaction times, responses, etc) by calling HCL_behavioural, and the associated “areas of interest” (described later) can be called with HCL_AOIs.

Counterbalanced designs

Many psychology experiments will counterbalance the position of important stimuli on the screen. In the example data, there are two stimuli, with one of these appearing on the left side of the screen and the other on the right. In the design of the experiment, one of these stimuli can be considered a “target” and the other a “distractor”, and the experiment counterbalances whether these are positioned in a left/right or a right/left arrangement across trials. In order to provide a meaningful analysis of the eye position over all trials, it is necessary to standardise the data, such that the resulting analyses reflect meaningful eye gaze on each type of stimulus (target or distractor).

eyetools has a built in function, `conditional_transform()`, which allows us to

transform the x and/or y values of the stimuli so as to take into account a counterbalancing variable. This function currently performs a single-dimensional transformation, across either the horizontal or vertical midline. It can be used on raw data or fixation data; we simply need to add a column to the data to reflect the counterbalancing variable. The result of the function is a set of data in which the x (and/or y) position is consistent across counterbalanced conditions (e.g., in our example, we can transform the data so that the target cue is always on the left). This transformation is especially useful for future visualisations and calculation of time on areas of interest.

In the example code, we have merged the eye data with a set of “trial_events” data that describe the events on each trial. We can apply `conditional_transform()` and specify the relevant column (`cue_order`) that controls the counterbalancing, and the relevant value that signals a switch of position (here the value “2”). By default the function expects a resolution of 1920x1080, but custom resolutions can be specified. The resulting transformation means that the data is normalised such that the target stimulus is always positioned on the left side of the screen.

```
# merges with the common variables pNum and trial
data <- merge(data, HCL_behavioural)

# perform a transformation of the data across the x coordinate midline
# for all trials with value 2 in the column cue_order
data <- conditional_transform(data,
                             flip = "x",
                             cond_column = "cue_order",
                             cond_values = "2")
```

Fixations

Once the data has been repaired and smoothed, a core step in eye data analysis is to identify fixations. Broadly, a fixation is defined as a period in which the eye stops moving and is held in a specific location for a significant period of time (typically longer than 100 ms; Salvucci

and Goldberg (2000)). The period in which the eyes are moving between fixations reflects a “saccade”. While the eyes move during these brief (typically less than 50 ms) periods of movement, significant perceptual suppression occurs and there is minimal information processing (Duren & Sanders, 1995; Irwin et al., 1995; Sanders & Houtmans, 1985). Therefore for many cognitive psychologists, the periods of fixation are particularly important and reflect the most relevant periods of information processing in a task.

The raw data can be transformed into these meaningful eye data characteristics. Beyond their importance for understanding psychological processes, transforming the data into fixations and saccades leads to greater computational efficiency. For example, the built in example data in *eyetools* is 479 kb, which contains 31,041 rows of data (constituting just 12 trials of data). After processing the data into fixations, the resulting data is 269 rows and can be saved as 3.8 kb, less than 1% the size of the raw data. Not only is this more computationally efficient, but it also means the data are now in a far more practical format for storage in online data repositories.

There are two fixation algorithms offered in the *eyetools* package, both based on methods presented by Salvucci and Goldberg (2000). The first, `fixation_dispersion()` seeks periods of low variability in the spatial component of the data; the algorithm looks for sufficient periods of time in which the gaze position remains within a tolerated maximum range of dispersion. Once this range is exceeded, this is deemed to be the end of a possible period of fixation. If the total time of this fixation period is longer than the minimum required (set by the `min_dur` parameter), then this fixation is stored as an entry in the returned object.

The second algorithm, `fixation_VTI()`, employs a velocity-threshold approach to identifying fixations, based on the algorithm described in Salvucci and Goldberg (2000). Since points of fixation occur when the eye is not in consistent motion, the algorithm computes the Euclidean distance between points and then determines the velocity of the eye. Periods in which this velocity is consistently below the velocity threshold (for which the default is 100 degrees of visual angle per second) are identified as a potential period of fixation. The algorithm then applies a dispersion check to ensure that the eye maintains a relatively stable position across this period.

Fixations must be of a minimum length for classification (by default 150 ms).

Here we can see the example data passed to the `fixation_dispersion()` algorithm and the resulting fixations that are returned.

```
fixations <-
  fixation_dispersion(data,
    min_dur = 150, # Min duration in ms
    disp_tol = 100, # Max dispersion tolerance in pixels
    NA_tol = 0.25, # proportion of NAs tolerated
    progress = FALSE) # toggle progress bar

head(fixations, 4)
```

pID	trial	fix_n	start	end	duration	x	y	prop_NA	min_dur	disp_tol
1 118	1	1	0	173	173	959	811	0	150	100
2 118	1	2	197	397	200	961	590	0	150	100
3 118	1	3	400	653	253	958	490	0	150	100
4 118	1	4	803	1083	280	1372	839	0	150	100

Saccades

Between periods of fixation, the velocity of the eye increases rapidly as it makes a saccade towards the next point of fixation. The `saccade_VTI()` function will extract saccades using the velocity threshold algorithm described above. The resulting output provides details of each saccade, such as the timing of the saccade onset, duration, and the origin and terminus coordinates. As with the fixation algorithms, default parameters have been chosen, but they can be adapted to fit the requirements of the researcher.

```
saccades <- saccade_VTI(data,
                        threshold = 150,
                        min_dur = 20)

head(saccades, 4)
```

	pID	trial	sac_n	start	end	duration	origin_x	origin_y	terminal_x	terminal_y
252										
253	1	118	1	1	2180 2240	60	833.2688	296.7871	487.3967	705.9158
254	2	118	1	2	2710 2750	40	614.5028	605.7001	862.3837	408.3421
255	3	118	1	3	3673 3726	53	885.6256	253.4150	558.1883	655.7776
256	4	118	1	4	4213 4233	20	460.3286	722.8386	577.2034	617.8567
257										
				mean_velocity	peak_velocity					
258	1			225.0736	331.8455					
259	2			200.3353	263.8863					
260	3			243.7927	340.3059					
261	4			195.6512	251.7763					

262 Area of interest (AOI) analysis

263 A critical component in many analyses of eye gaze is the assessment of time spent in
 264 regions of space. *eyetools* has a number of functions for assessing the time spent in Areas of
 265 Interest (AOIs), as well as the sequence in which the eye enters and exits these areas. AOIs will
 266 typically reflect regions of space in which critical stimuli appear. AOIs are defined in *eyetools*
 267 using a dataframe object, where each row reflects a unique AOI, where values code for the
 268 centrepoint of the AOI in x/y coordinates along with the width and height (if the AOIs are
 269 rectangular) or just the radius (if circular). This object can be created using the function
 270 `create_AOI_df()`:

```
# set areas of interest
AOI_areas <- create_AOI_df(3)

# populate this dataframe with AOI dimensions
# (x, y, width/radius, height)
AOI_areas[1,] <- c(460, 840, 400, 300) # Left rectangular AOI
AOI_areas[2,] <- c(1460, 840, 200, NA) # Right circular AOI
AOI_areas[3,] <- c(960, 840, 200, 400) # Centre rectangular AOI
```

271 The function `AOI_time()` function calculates the time spent in each AOI per trial, using
 272 either the raw data or fixation data as input. The resulting output can be expressed in the form of
 273 absolute time, or, by passing a vector of times to the “`trial_time`” parameter, can be expressed as
 274 proportional time.

```
data_AOI_time <-
  AOI_time(data = fixations,
           data_type = "fix",
           AOIs = HCL_AOIs,
           AOI_names = c("target", "distractor", "outcomes"),
           as_prop = TRUE,
           trial_time = HCL_behavioural$RT)

head(data_AOI_time, 9)
```

	pID	trial	AOI	time
275	1	118	1	target
276				0.1043446
277	2	118	1	distractor
				0.2488674
278	3	118	1	outcomes
				0.4041589


```

279 4 118      2      target 0.1380248
280 5 118      2 distractor 0.1702220
281 6 118      2    outcomes 0.4816758
282 7 118      3      target 0.1391737
283 8 118      3 distractor 0.1080352
284 9 118      3    outcomes 0.5397965

```

285 We can see from the resulting data that the function provides time on each AOI for each
 286 trial. Used in combination with the `conditional_transform()` function, `AOI_time()` provides
 287 a very efficient way to assess time on critical regions of space. Since the data is in long format, it
 288 can be easily processed further with common techniques in R:

```

library(dplyr)

data_AOI_time %>%
  dplyr::group_by(AOI) %>%
  dplyr::summarise(mean_time = mean(time))

```

```

289 # A tibble: 3 x 2
290   AOI      mean_time
291   <fct>      <dbl>
292 1 target      0.201
293 2 distractor  0.237
294 3 outcomes    0.365

```

295 The `AOI_time_binned()` function can assess the duration of time spent in AOIs, divided
 296 into sequential time bins. Since fixations will naturally overlap these segments in many
 297 circumstances, this function operates only on raw data. Here we are assessing time in the three
 298 AOIs for periods of 1000 ms in length, and limiting this analysis to the first 8000 ms.

```

data_AOI_time_binned <-
  AOI_time_binned(data,
    AOIs = HCL_AOIs,
    AOI_names = c("target", "distractor", "outcomes"),
    bin_length = 1000, # in milliseconds
    max_time = 8000) # in milliseconds

head(data_AOI_time_binned, 10)

```

	pID	trial	bin_n	target	distractor	outcomes
299						
300	1	118	1	1	0	217
301	2	118	1	2	0	187
302	3	118	1	3	460	0
303	4	118	1	4	270	0
304	5	118	1	5	220	0
305	6	118	1	6	0	630
306	7	118	1	7	0	797
307	8	118	1	8	0	370
308	9	118	2	1	617	0
309	10	118	2	2	167	0

310 It is also possible to determine the sequence of entries into AOIs using the `AOI_seq()`
 311 function. This function currently works only with fixation data. For a given trial, the sequence of
 312 fixations is assessed against the AOIs provided, where consecutive fixations within the same AOI
 313 are combined into one “entry period”. The result of this function is a sequence of AOI entries per
 314 trial for each participant, providing data on the sampling order of AOIs. The resulting output
 315 provides start and end times and duration of each entry.

```
data_AOI_entry <-
  AOI_seq(fixations,
          AOIs = HCL_AOIs,
          AOI_names = c("target", "distractor", "outcomes"))

head(data_AOI_entry, 9)
```

	pID	trial	AOI	start	end	duration	entry_n
316							
317	1	118	1	outcomes	400	653	253
318	2	118	1	distractor	803	1083	280
319	3	118	1	outcomes	1233	2120	887
320	4	118	1	target	2260	2666	406
321	5	118	1	outcomes	2760	3646	886
322	6	118	1	target	3753	4116	363
323	7	118	1	outcomes	4286	5323	1037
324	8	118	1	distractor	5403	6772	1369
325	9	118	1	distractor	7652	9272	1620

326 Knowing the order in which the eyes visit particular regions of space is essential for many
 327 steps in eye data analysis. For example, each trial might start with a fixation point in the centre of
 328 the screen. Below we show how the `AOI_seq()` function can be used in combination with other
 329 basic R commands to efficiently detect the first AOI entry on each trial. We can see that in all but
 330 one of the 12 example trials, participants process the central fixation point first.

```
library(dplyr)

# add a central fixation AOI region
HCL_AOIs[4,] <- c(960, 810, 200, 200)
```

```

data_AOI_entry <-
  AOI_seq(fixations,
          AOIs = HCL_AOIs,
          AOI_names = c("target", "distractor", "outcomes", "fixation"))

data_AOI_entry %>%
  dplyr::group_by(pID, trial) %>%
  dplyr::slice(1)

```

```

331 # A tibble: 12 x 7
332 # Groups:   pID, trial [12]
333   pID   trial AOI      start   end duration entry_n
334   <chr> <dbl> <chr>    <dbl> <dbl>    <dbl>    <dbl>
335 1 118     1 fixation    0   173     173      1
336 2 118     2 fixation    0   186     186      1
337 3 118     3 outcomes 393   906     513      1
338 4 118     4 fixation    0   153     153      1
339 5 118     5 fixation   10   163     153      1
340 6 118     6 fixation    0   177     177      1
341 7 119     1 fixation    0   246     246      1
342 8 119     2 fixation    0   160     160      1
343 9 119     3 fixation    0   180     180      1
344 10 119    4 fixation    0   227     227      1
345 11 119    5 fixation    0   197     197      1
346 12 119    6 fixation    0   260     260      1

```

347 Visualisations

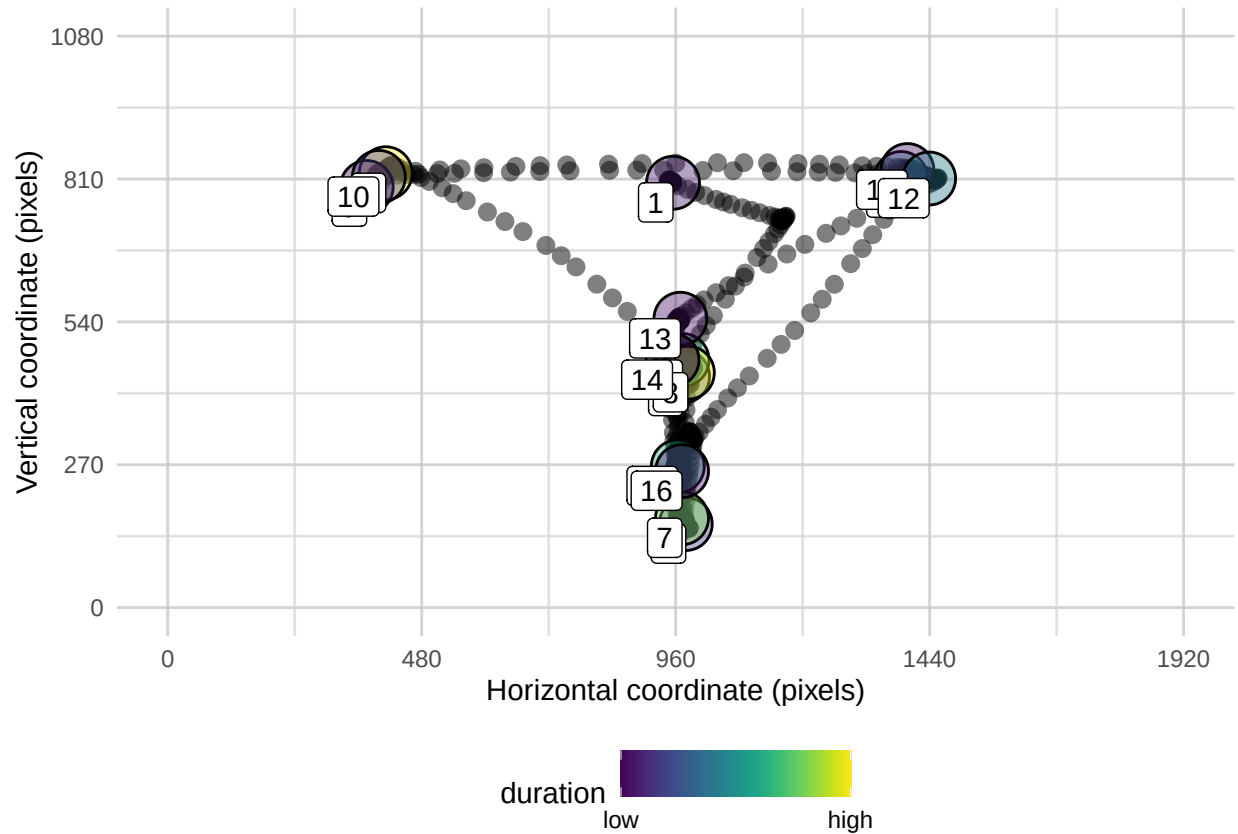
348 The *eyetools* package has a number of built in visualisations that allow for functional plots
349 of the data, with minimal effort. All plots use the dominant graphical R package *ggplot*, which
350 means that the resulting plots from these functions are *ggplot* objects and can therefore be
351 customised using the full suite of options for *ggplot* and its extensions.

352 `plot_spatial()` offers a simple means to view the data produced by *eyetools*. By default
353 this will plot all of the data that is passed to the function, but participant IDs and trial values can
354 be specified in order to plot specific data. Here we plot the raw data from a single trial for one
355 participant, with the detected fixations overlaid. When using fixation data, the fixations are
356 labelled in their temporal order (by default), enabling a clear presentation of how the fixations
357 arose.

```
plot_spatial(raw_data = data,  
             fix_data = fixations,  
             pID_values = 118,  
             trial_values = 6)
```

Figure 1

Raw data (grey points) and fixations (coloured circles) for a single trial

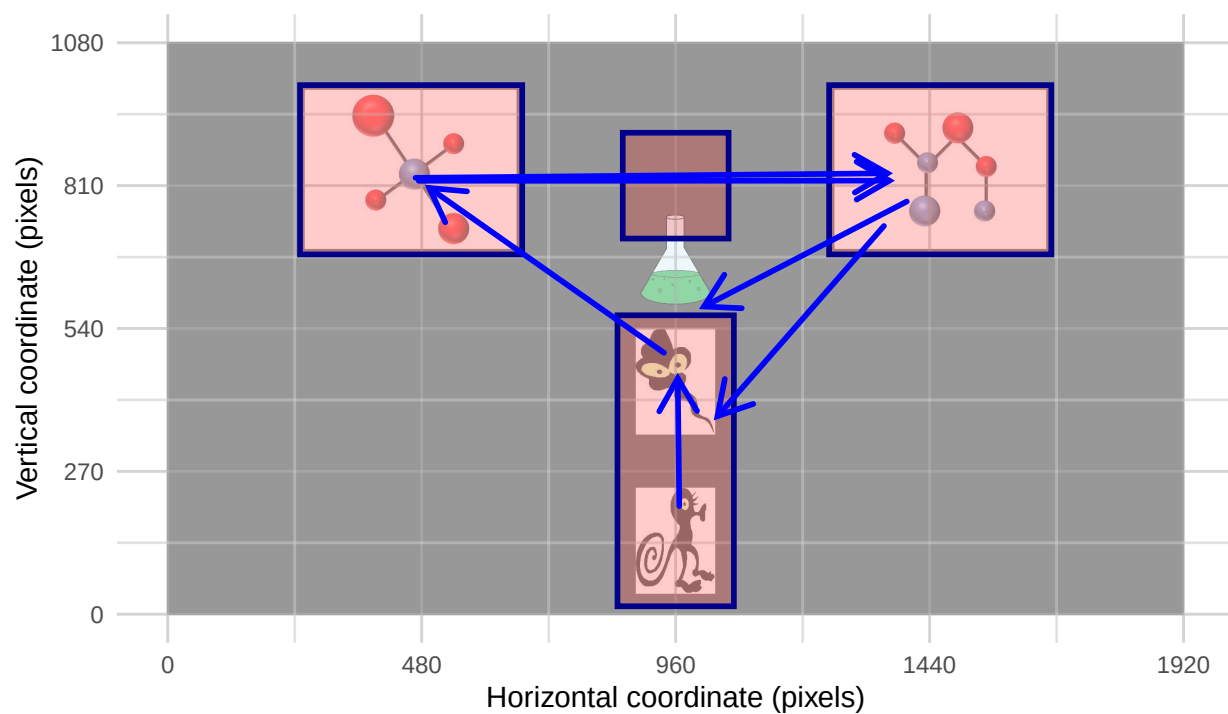


In addition to eye data, a background image can be added to the plot, which is useful for inspecting data over a representation of the experimental task. If AOIs have been defined, these can be plotted as well. Here we demonstrate the plotting of the saccades, AOIs, and a background image:

```
plot_spatial(sac_data = saccades,  
             AOIs = HCL_AOIs,  
             pID_values = 118,  
             trial_values = 6,  
             bg_image = "images/HCL_sample_image.png")
```

Figure 2

Saccades (blue arrows) and Area-Of-Interest regions (pink shapes) for a single trial, against a background image.



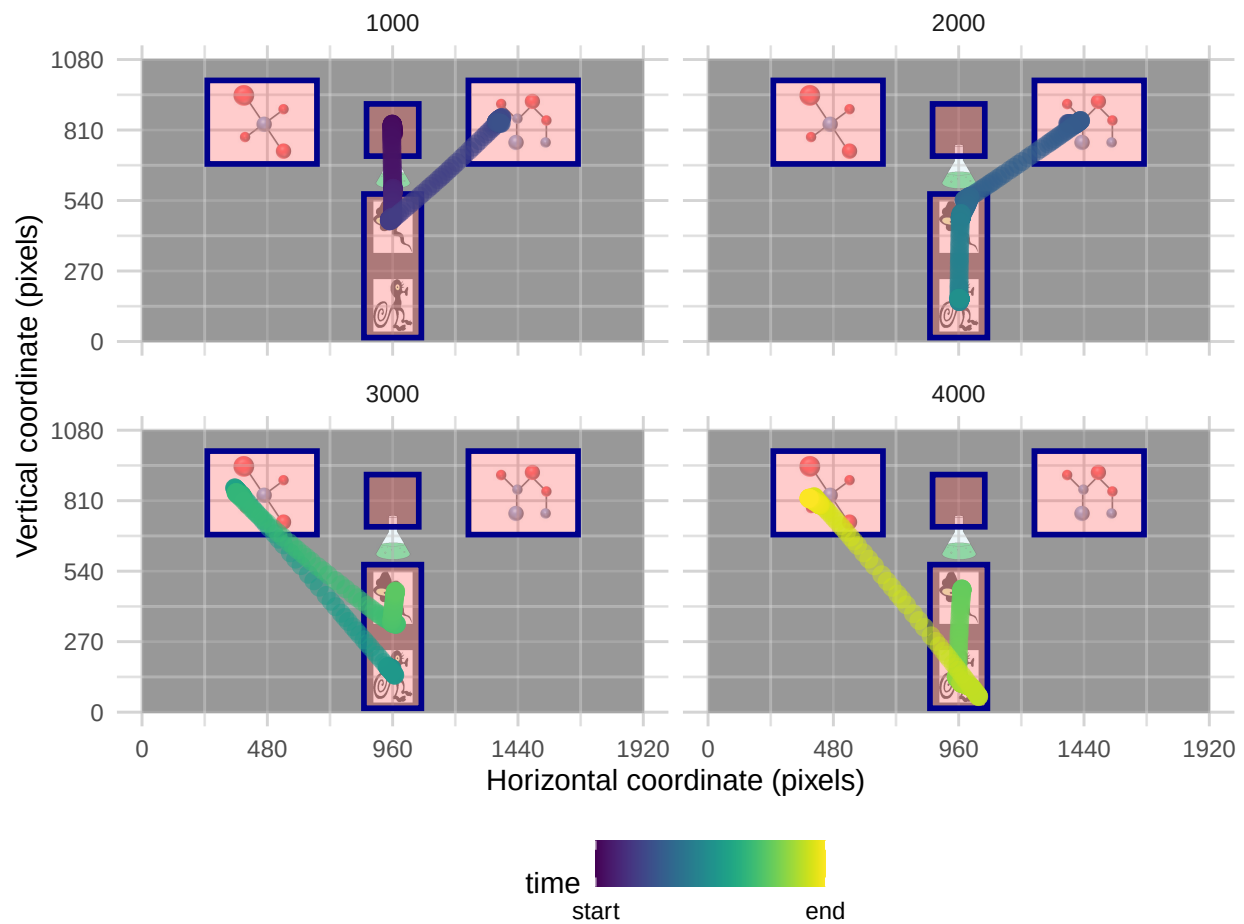
The function `plot_seq()` is useful for visualising data as a series of plots, mapping out eye movements over the course of a single trial. By default this function will plot a randomly selected trial from the raw data that is passed to the function. Otherwise, specific trials and participant values can be specified. The function requires a “bin_time” parameter, that specifies

366 the length of each time-period within the trial. An optional parameter of “bin_range” can be
367 specified to restrict the range of these periods that are presented. For example here we plot data in
368 periods of 1000 ms across the first four of these periods.

```
plot_seq(data = data,  
         bin_time = 1000,  
         bin_range = c(1,4),  
         trial_values = 1,  
         pID_values = 118,  
         AOIs = HCL_AOIs,  
         bg_image = "images/HCL_sample_image.png")
```

Figure 3

The data from the same example trial, plotted across 4 consecutive bins of 1000 milliseconds



The `plot_AOI_growth()` function offers a visualisation of the progression of time spent on AOIs across a single trial. This can be useful to see how participants interact with AOIs over time, and this can be presented as either a plot of the cumulative time, or as a proportion of the time spent in the trial.

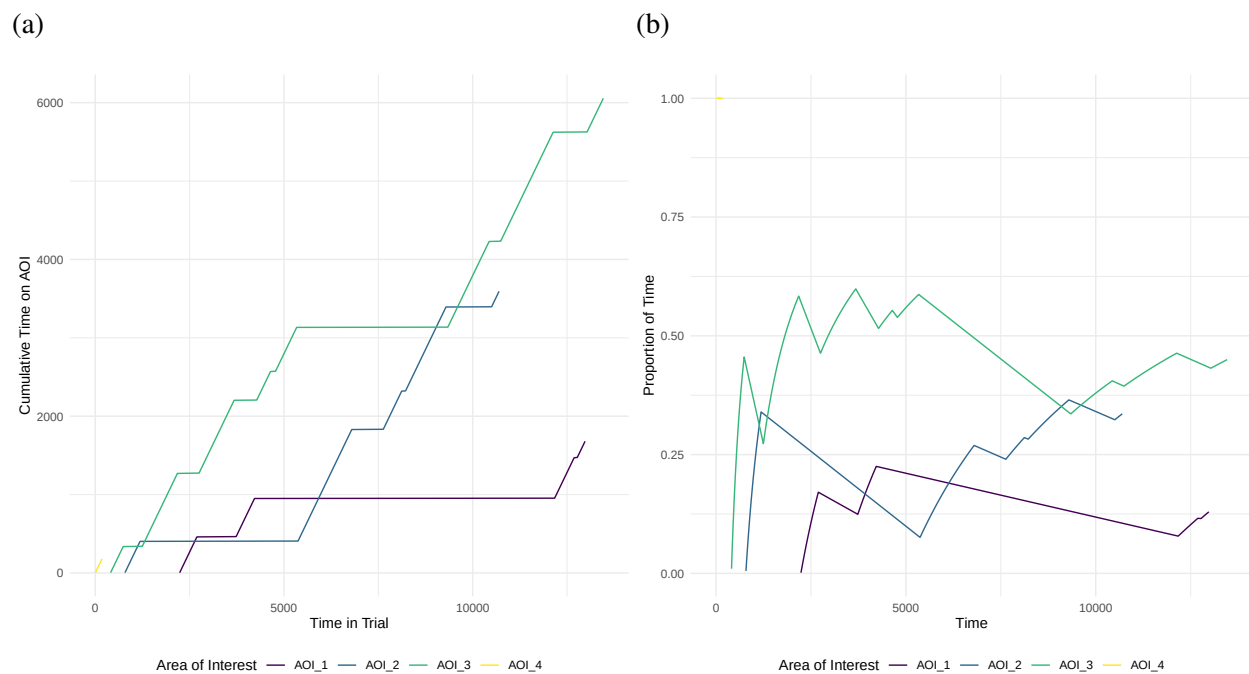
```
# plot absolute and then proportional

plot_AOI_growth(data = data,
                AOIs = HCL_AOIs,
                type = "abs",
                pID_values = 118,
                trial_values = 1)

plot_AOI_growth(data = data,
                AOIs = HCL_AOIs,
                type = "prop",
                pID_values = 118,
                trial_values = 1)
```

Figure 4

Examples of the absolute and proportional time plots from `plot_AOI_growth()`



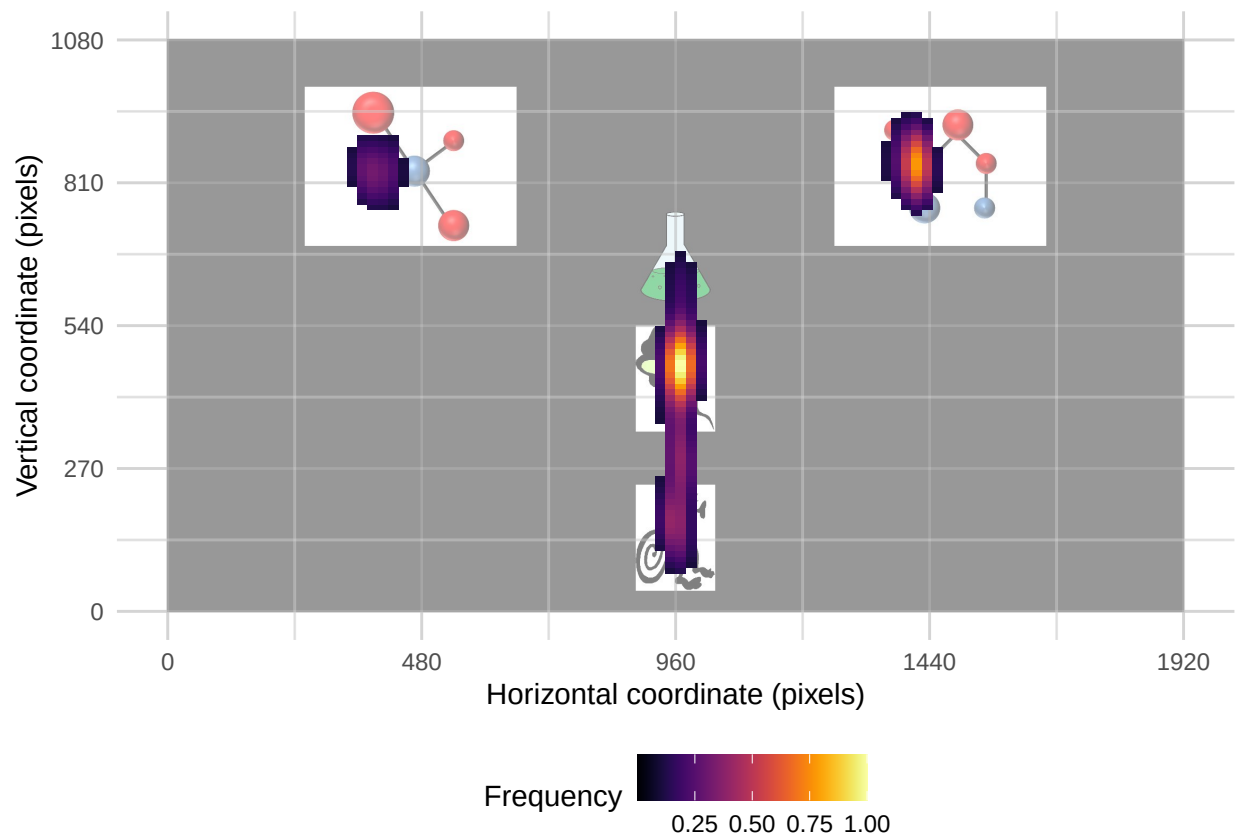
A heatmap of eye gaze positions can be generated using `plot_heatmap()` which takes raw data as input. Like `plot_spatial()`, it is possible to select certain `pID` and `trial_values`,

therefore offering a complementary visualisation of raw data. As can be seen in Figure 5, we can be reassured that participants do indeed spend most of their time looking at the stimuli on screen rather than in the empty space. `plot_heatmap()` also allows for the modification of the amount of data displayed, using the `alpha_range` parameter. This takes a pair of values to specify a range between 0 and 1. The first value is a cut off, controlling how much of the low frequency positions are not displayed in the plot. The second value sets the transparency of the visible points.

```
plot_heatmap(data,  
              pID_values = 118,  
              trial_values = c(1,3),  
              alpha_range = c(0.1,1),  
              bg_image = "images/HCL_sample_image.png")
```

Figure 5

A heatmap overlaid upon a sample stimuli image demonstrating where the participants looked most over all trials



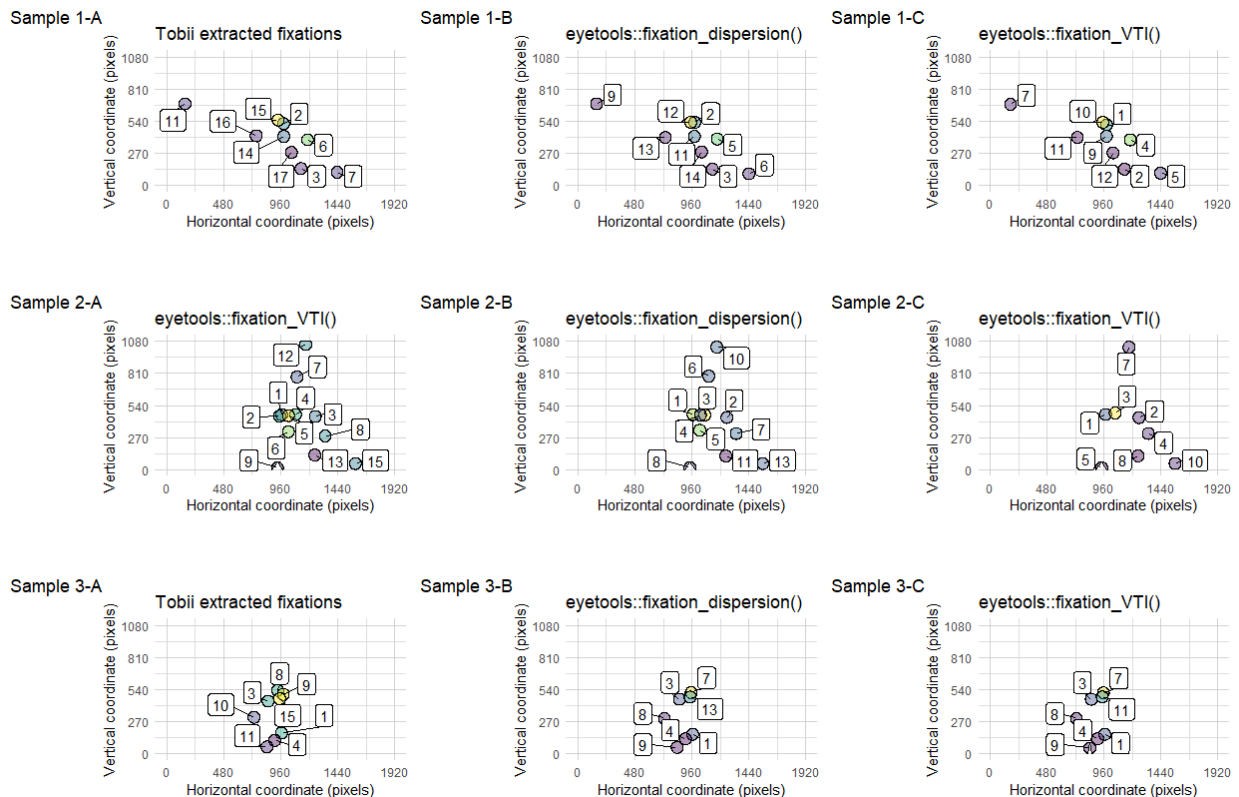
Validation of fixation metrics

eyetools uses implementations of common methods for extracting fixations from raw data (Salvucci & Goldberg, 2000). To provide a simple validation of our primary fixation algorithm,

we took raw data collected independently from a researcher outside of our lab on an unknown task. The researcher provided the raw data and extracted fixations for a 6 minute period of data collection, from Tobii Pro Lab software. From the raw data we used the *eyetools* `fixation_dispersion()` and `fixation_VTI()` algorithms to compute fixations from randomly drawn periods of 10 seconds. Figure 6 shows side-by-side comparisons of Tobii and *eyetools* extracted fixations from 3 such periods (the raw data and analysis script for this comparison is available in the manuscript repository for full exploration of other periods). Somewhat unsurprisingly, the algorithms show a very similar spread of fixations for these periods. Notably the number of overall fixations differs across the samples, which is a consequence of the particular parameters used to define fixations, such as the dispersion tolerance and the minimum duration.

Figure 6

A comparison of extracted fixations from Tobii Pro Lab (A:left), eyetools::fixation_dispersion() (B:centre) and eyetools::fixation_VTI() (C:right), across 3 samples.



Summary and future directions

This paper has given an introduction and basic tutorial on working with an open source R package for an analysis pipeline for eye data. We began by identifying the current gaps in available tools for working with eye data in reproducible pipelines. We then provided an overview of the initial steps in working with raw eye data and the conversion of raw eye data into a usable format for working with the functions in *eyetools*. We then covered many useful steps in the analysis pipeline using functions available in the *eyetools* package that included the repairing and normalising of the data, the detection of events such as fixations and saccades, and the trial level analysis of data patterns, such as time on areas of interest, and the sequencing of entries to areas of interest.

This tutorial offers a step-by-step walk-through for handling eye data using R, demonstrating that the *eyetools* package provides a set of tools that will lead to reproducible analysis steps for many experimental psychologists. It is hopefully clear that the functions in *eyetools* are flexible and powerful, yet ultimately simple to implement. While these functions represent some initial steps in eye data analysis, since the objects that result from these functions are standard formats in R (i.e., dataframes and plots), they will provide the user a means to enable more complex or nuanced analyses.

eyetools offers an open-source toolset that holds no hidden nor proprietary functionality. The major benefits of open-source tools are extensive: not only do they allow for full inspection and reproducibility of analyses, but they also support and enable the development and sharing of new analysis functions. There are a number of obvious features that would be hugely beneficial in future versions of *eyetools*. For example, while the package provides means to determine the order of eye-movements (`AOI_seq()`) there is no means to compare these patterns across different trials or periods. Such “scan-path analyses” have been used effectively in cognitive studies and so algorithms to conduct these analyses would be an obvious next step. There are also no functions within the package to handle pupilometry data, despite the obvious benefits of analysing these data. From a development perspective, while these features would be hugely beneficial to the

package, they will only be implemented as and when there is a need in our research. Thus our hope for *eyetools* is that future versions will benefit from user engagement and an expansion of the toolset to enable an ever more powerful set of features. We believe *eyetools* provides a solid foundation for this collaborative venture.

Data and code availability

This manuscript was written in Quarto and can be reproduced from the manuscript source files which are available at https://github.com/tombeasley/BRM_eyetools . The manuscript details functions from the latest development version of *eyetools* which is 0.9.3. We welcome contributions to the development of *eyetools* by posting bug reports and suggested improvements at <https://github.com/tombeasley/eyetools/issues> .

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Availability of data and materials

This manuscript was written in Quarto and can be reproduced from the manuscript source files which are available at https://github.com/tombeasley/BRM_eyetools

Code availability

The manuscript details functions from the latest CRAN version of *eyetools* which is 0.9.2. The package source code is available at <https://github.com/tombeesley/eyetools>

Authors' contributions

TB - project inception; software design and coding; primary author of the manuscript. MI - software design and coding; secondary author of the manuscript

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